2. Fireworks contain a range of chemicals including a fuel, oxidising agents and metal salts.

(a) One oxidising agent used in fireworks is potassium perchlorate, $KClO_4$. This reacts with aluminium metal and produces a bright flash.

The equation for the reaction is

$$KClO_4 \quad + \quad Al \quad \rightarrow \quad KCl \quad + \quad Al_2O_3$$

Balance this equation. **1**

(b) Fireworks were traditionally made using compounds containing the chlorate ion, ClO_3^-, as an oxidising agent.

(i) Chlorate ions release oxygen when they decompose.

Potassium chlorate, $KClO_3$, ($GFM = 122.6$ g) reacts as shown.

$$2KClO_3(s) \rightarrow 3O_2(g) + 2KCl(s)$$

Calculate the volume of oxygen produced, in litres, when 4.6 g of potassium chlorate decomposes. **2**

Take the volume of 1 mole of oxygen gas to be 24 litres.

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2. (b) (continued)

(ii) The decomposition of potassium chlorate can be speeded up by the addition of a catalyst.

State the effect of adding a catalyst on the enthalpy change for this reaction. **1**

(iii) A firework containing 5.5 g of potassium perchlorate ($GFM = 138.6$ g) releases 103 kJ of energy.

Calculate the energy, in kJ, released per mole of potassium perchlorate. **1**

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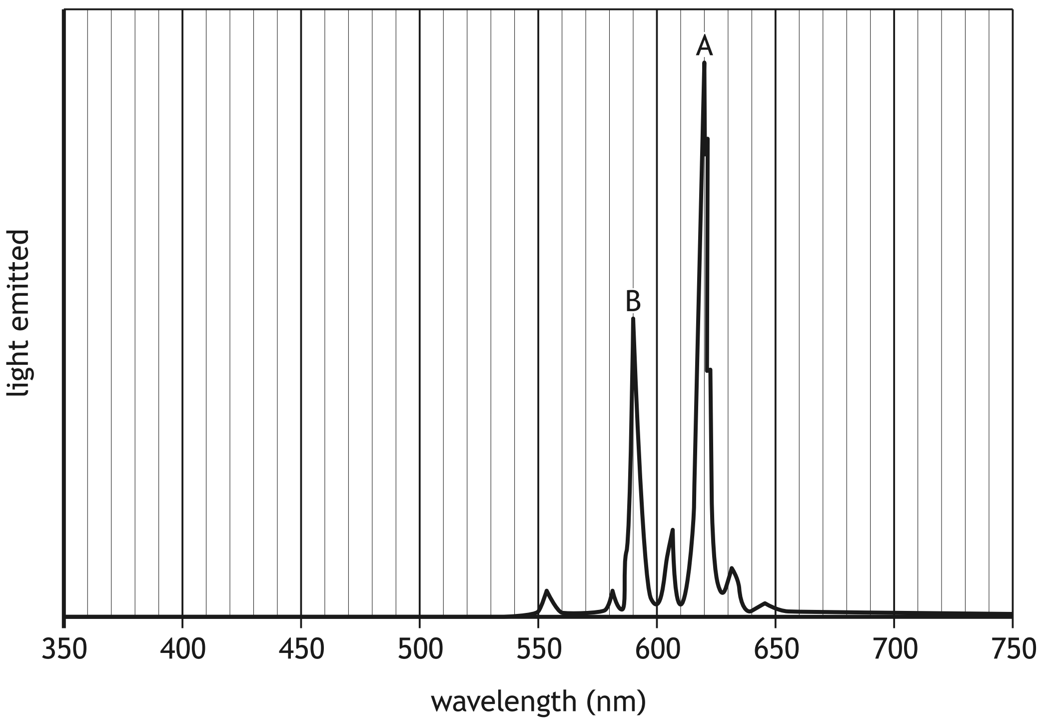
2. (b) (continued)

(iv) Explain **fully** why increasing temperature increases the rate of a chemical reaction. **2**

(c) The different flame colours produced by metal salts are caused by different wavelengths of light being emitted. The flame colours associated with different wavelengths are given in the data booklet.

The following profile shows the colours emitted by one particular firework. Each peak represents a different colour of light.



Peak A has a wavelength of 620 nm, corresponding to red light.

Suggest the **metal** responsible for peak B on the spectrum. **1**

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